

APPLICATION NOTE

Linux 通过配置文件调节发射功率

**ATBM6162**

1x1
802.11b/g/n
/ac/ax
Wi-Fi 芯片

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Linux 系统下 ATBM WIFI P2P 使用指南

作者	版本	说明
傅贵	V1.0	init

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1 按频段配置功率

1.1 按照不同频段配置功率

命令: iwpriv wlan0 common setttxpower_byfile,1

参数: 1: 使用配置文件中的 deltagain; 0: 使用芯片 efuse 中的 deltagain;

注意: 必须在发包之后使用 setttxpower_byfile 命令才能生效

(1) 说明

Atbm wifi 驱动目前支持加载驱动时读取配置文件来调节当前发射功率, 功率调节分别满足高中低频点配置文件的存放位置及文件名可以通过 atbm_wifi_driver\hal_apollo 目录下的 Makefile 来进行配置:

```
16: #####
17: #ccflags-y += -DCONFIG_ATBM_APOLLO_5GHZ_SUPPORT
18: #ccflags-y += -DCONFIG_ATBM_APOLLO_WAPI_SUPPORT
19: #ccflags-y += -I$(shell pwd)/../include/ \
20:     -include $(shell pwd)/../include/linux/compat-2.6.h \
21:     -DCOMPAT_STATIC
22: ##指定配置文件路径, 当配置文件存在时使用配置文件中的dcxo值不使用efuse中的dcxo, 配置文件中的delta_gain为修i
23: ATBM_CONFIG_FILE="/tmp/atbm_txpwr_dcxo_cfg.txt"
24: ccflags-y += -DCONFIG_TXPOWER_DCXO_VALUE="\$(ATBM_CONFIG_FILE)\\"
25: # use the deltagain value in file :"\$(ATBM_CONFIG_FILE)",when file exist
26: ATBM_RATE_POWER_CONFIG_FILE="/tmp/set_rate_power.txt"
27: ccflags-y += -DCONFIG_RATE_TXPOWER="\$(ATBM_RATE_POWER_CONFIG_FILE)\\"
28:
29: ifneq ($(TF1NAMEF).)
```

文件内容格式如下:

格式要求: **必须按照如下格式配置, 打乱顺序会导致配置错位。**

```
delta_gain1:31
delta_gain2:0
delta_gain3:1
delta_gain1_5g:-31
delta_gain2_5g:-31
delta_gain3_5g:-31
delta_gain4_5g:-11
delta_gain5_5g:31
delta_gain6_5g:31
delta_gain7_5g:31
delta_gain8_5g:31
delta_gain9_5g:31
delta_gain10_5g:-1
dcxo:75
```

数据有效范围 (即调节范围 $[-7.75\text{dB} \sim 7.75\text{dB}]$):

名称	有效范围	调节范围	物理意义
delta_gain1	-31~31	信道 1~信道 4	$(\text{delta_gain1}/4)\text{dB}$

delta_gain2	-31~31	信道 5~信道 9	(delta_gain2/4)dB
delta_gain3	-31~31	信道 10~信道 14	(delta_gain3/4)dB
delta_gain1_5g	-31~31	信道 36~信道 40	(delta_gain1_5g/4)dB
delta_gain2_5g	-31~31	信道 44~信道 52	(delta_gain2_5g/4)dB
delta_gain3_5g	-31~31	信道 56~信道 64	(delta_gain3_5g/4)dB
delta_gain4_5g	-31~31	信道 100~信道 104	(delta_gain4_5g/4)dB
delta_gain5_5g	-31~31	信道 108~信道 116	(delta_gain5_5g/4)dB
delta_gain6_5g	-31~31	信道 120~信道 128	(delta_gain6_5g/4)dB
delta_gain7_5g	-31~31	信道 132~信道 140	(delta_gain7_5g/4)dB
delta_gain8_5g	-31~31	信道 144~信道 149	(delta_gain8_5g/4)dB
delta_gain9_5g	-31~31	信道 153~信道 157	(delta_gain9_5g/4)dB
delta_gain10_5g	-31~31	信道 161~信道 165	(delta_gain10_5g/4)dB
dcxo	0~127	频偏	无

(2) 配置方式

在芯片 efuse 原有 deltagain 的基础上做补偿

注意

数据都不会写到 efuse , 该配置掉电就没了, 不会保存到 ROM

PS:

驱动在加载的时候会默认去读取这个文件并根据配置文件内容生效。

运行过程中修改了配置文件内容, 并想立即生效直接执行命令:

```
iwpriv wlan0 common settxpower_byfile
```

驱动会去刷新配置并立即生效。

2 按速率配置功率

2.1 修改不通速率的功率

命令：iwpriv wlan0 common set_rate_power,<useflag>

useflag:是否使用从配置文件中读取的数据，1：使用；0：使用默认发送功率

(1) 配置方式

配置文件的存放位置及文件名可以通过 atbm_wifi_driver\hal_apollo 目录下的 Makefile 来进行配置：

```
186 # use the deltagain value in file : "$(ATBM_CONFIG_FILE)", when file exist
187 ATBM_RATE_POWER_CONFIG_FILE="/tmp/set_rate_power.txt"
188 ccflags-y += -DCONFIG_RATE_TXPOWER="\$(ATBM_RATE_POWER_CONFIG_FILE)\\"
189
190 ATBM_RATE_POWER_CONFIG_FILE_5G="/tmp/set_rate_power_5g.txt"
191 ccflags-y += -DCONFIG_RATE_TXPOWER_5G="\$(ATBM_RATE_POWER_CONFIG_FILE_5G)\\"
192
```

2.4G 配置文件格式：

```
target_or_delta=1
b_1M_2M=210
b_5_5M_11M=210
g_6M_n_6_5M=210
g_9M=200
g_12M_n_13M=190
g_18M_n_19_5M=190
g_24M_n_26M=190
g_36M_n_39M=190
g_48M_n_52M=190
g_54M_n_58_5M=180
n_65M=180
VHT_MCS0=210
VHT_MCS1=190
VHT_MCS2=190
VHT_MCS3=190
VHT_MCS4=190
VHT_MCS5=190
VHT_MCS6=180
VHT_MCS7=170
VHT_MCS8=170
VHT_MCS9=170
```

HE-SU_MCS0=210
HE-SU_MCS1=190
HE-SU_MCS2=190
HE-SU_MCS3=190
HE-SU_MCS4=190
HE-SU_MCS5=190
HE-SU_MCS6=180
HE-SU_MCS7=180
HE-SU_MCS8=170
HE-SU_MCS9=170
HE-SU_MCS10=160
HE-SU_MCS11=160
g_6M_n_6_5M_40M=200
g_9M_40M=190
g_12M_n_13M_40M=190
g_18M_n_19_5M_40M=190
g_24M_n_26M_40M=190
g_36M_n_39M_40M=190
g_48M_n_52M_40M=190
g_54M_n_58_5M_40M=180
n_65M_40M=170
VHT_MCS0_40M=200
VHT_MCS1_40M=190
VHT_MCS2_40M=190
VHT_MCS3_40M=190
VHT_MCS4_40M=190
VHT_MCS5_40M=190
VHT_MCS6_40M=180
VHT_MCS7_40M=170
VHT_MCS8_40M=170
VHT_MCS9_40M=170
HE-SU_MCS0_40M=200
HE-SU_MCS1_40M=190
HE-SU_MCS2_40M=190
HE-SU_MCS3_40M=190
HE-SU_MCS4_40M=190
HE-SU_MCS5_40M=190
HE-SU_MCS6_40M=180
HE-SU_MCS7_40M=170
HE-SU_MCS8_40M=170
HE-SU_MCS9_40M=170
HE-SU_MCS10_40M=160
HE-SU_MCS11_40M=160

5G 配置文件格式:

```
target_or_delta=1
g_6M_n_6_5M=200
g_9M=190
g_12M_n_13M=190
g_18M_n_19_5M=180
g_24M_n_26M=180
g_36M_n_39M=170
g_48M_n_52M=170
g_54M_n_58_5M=160
n_65M=160
VHT_MCS0=200
VHT_MCS1=190
VHT_MCS2=180
VHT_MCS3=180
VHT_MCS4=170
VHT_MCS5=170
VHT_MCS6=160
VHT_MCS7=160
VHT_MCS8=160
VHT_MCS9=150
HE-SU_MCS0=200
HE-SU_MCS1=190
HE-SU_MCS2=180
HE-SU_MCS3=180
HE-SU_MCS4=170
HE-SU_MCS5=170
HE-SU_MCS6=160
HE-SU_MCS7=160
HE-SU_MCS8=160
HE-SU_MCS9=150
HE-SU_MCS10=140
HE-SU_MCS11=130
g_6M_n_6_5M_40M=190
g_9M_40M=170
g_12M_n_13M_40M=170
g_18M_n_19_5M_40M=160
g_24M_n_26M_40M=160
g_36M_n_39M_40M=150
g_48M_n_52M_40M=150
g_54M_n_58_5M_40M=140
n_65M_40M=130
```

```
VHT_MCS0_40M=190
VHT_MCS1_40M=170
VHT_MCS2_40M=160
VHT_MCS3_40M=160
VHT_MCS4_40M=150
VHT_MCS5_40M=150
VHT_MCS6_40M=140
VHT_MCS7_40M=130
VHT_MCS8_40M=110
VHT_MCS9_40M=110
HE-SU_MCS0_40M=190
HE-SU_MCS1_40M=170
HE-SU_MCS2_40M=160
HE-SU_MCS3_40M=160
HE-SU_MCS4_40M=150
HE-SU_MCS5_40M=150
HE-SU_MCS6_40M=140
HE-SU_MCS7_40M=130
HE-SU_MCS8_40M=110
HE-SU_MCS9_40M=110
HE-SU_MCS10_40M=80
HE-SU_MCS11_40M=80
```

如果不使用 HT40 模式直接将这部分的功率配置直接写 0 即可。

target_or_delta 只能配置为 1，调整绝对发送功率。

示例：

target_or_delta=1:

数据配置后仍应该使得发送功率在 20dbm 以内，配置的数据是 10*dB。

如果设置 b_1M_2M=180，则 1/2M 速率发射功率为 18dBm。

如果设置 g_6M_n_6_5M=140，则 6M/6.5M 速率发射功率为 14dBm。

(2) 注意

- 1) 驱动在加载的时候默认去读取这个文件并根据配置文件内容生效。

运行过程中修改了配置文件内容，并想立即生效直接执行命令：

```
iwpriv wlan0 common set_rate_power,1
```

驱动会去刷新配置并立即生效。

使用命令将设置好的配置复位：

```
iwpriv wlan0 common set_rate_power,0
```

3 获取当前的配置信息

3.1 通过 iwpriv 方式

命令格式为：

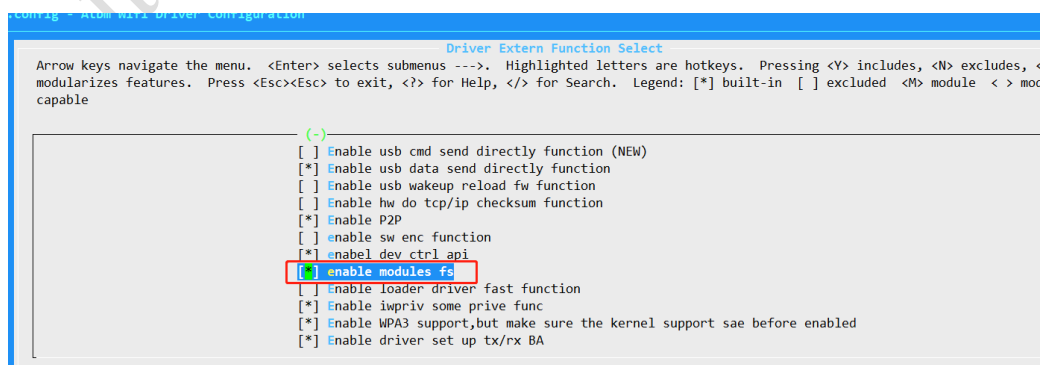
```
iwpriv wlan0 common get_cfg_txpower
```

```
[atom_firmware] HE-SU_MCS11_40M=0
wlan0 common:b_1M_2M=0
b_5_5M_11M=0
g_6M_n_6_5M=0
g_9M=0
g_12M_n_13M=0
g_18M_n_19_5M=0
g_24M_n_26M=0
g_36M_n_39M=0
g_48M_n_52M=0
g_54M_n_58_5M=0
n_65M=0
HE-SU_MCS0=0
HE-SU_MCS1=0
HE-SU_MCS2=0
HE-SU_MCS3=0
HE-SU_MCS4=0
HE-SU_MCS5=0
HE-SU_MCS6=0
HE-SU_MCS7=0
HE-SU_MCS8=0
HE-SU_MCS9=0
HE-SU_MCS10=0
HE-SU_MCS11=0
g_6M_n_6_5M_40M=0
g_9M_40M=0
g_12M_n_13M_40M=0
g_18M_n_19_5M_40M=0
g_24M_n_26M_40M=0
g_36M_n_39M_40M=0
g_48M_n_52M_40M=0
g_54M_n_58_5M_40M=0
n_65M_40M=0
HE-SU_MCS0_40M=0
HE-SU_MCS1_40M=0
HE-SU_MCS2_40M=0
HE-SU_MCS3_40M=0
HE-SU_MCS4_40M=0
HE-SU_MCS5_40M=0
HE-SU_MCS6_40M=0
HE-SU_MCS7_40M=0
HE-SU_MCS8_40M=0
HE-SU_MCS9_40M=0
HE-SU_MCS10_40M=0
HE-SU_MCS11_40M=0

delta_gain1:27 delta_gain2:27 delta_gain3:27 dcxo:50
```

3.2 通过 module_fs 方式获取

需要 ATBM_WIFI 驱动打开 MODULES_FS 宏



```
cat /sys/module/<MODULES_NAME>/atbmfs/ atbm_get_cfg_power
```

当前距离驱动的 MODULES_NAME 为:

```
/sys/module/ATBM606x_wifi_usb/atbmfs # lsmod
ATBM606x_wifi_usb 626103 0 - Live 0xbfc55000 (o)
cfg80211 285288 1 ATBM606x_wifi_usb, Live 0xbfb74000 (o)
nls_utf8 1380 0 - Live 0xbfb70000
exfat 64522 0 - Live 0xbfb59000
drv_ms_cus_gc2053_MIP 6865 0 - Live 0xbfb53000 (PO)
mi_shadow 34439 0 - Live 0xbfb45000 (PO)
mi_venc 165330 0 - Live 0xbfb12000 (PO)
mi_vif 61952 0 - Live 0xbfafc000 (PO)
mi_sensor 22016 0 - Live 0xbfaf2000 (PO)
mi_vpe 174963 0 - Live 0xbfabd000 (PO)
mi_drv 48628 0 - Live 0xbfab5000 (PO)
```

所以整体的执行命令为:

```
cat /sys/module/ ATBM606x_wifi_usb/atbmfs/ atbm_get_cfg_power
```

```
/tmp # cat /sys/module/ATBM606x_wifi_usb/atbmfs/atbm_get_cfg_power
b_1M_2M=0
b_5_5M_11M=0
g_6M_n_6_5M=0
g_9M=0
g_12M_n_13M=0
g_18M_n_19_5M=0
g_24M_n_26M=0
g_36M_n_39M=0
g_48M_n_52M=0
g_54M_n_58_5M=0
n_65M=0
HE-SU_MCS0=0
HE-SU_MCS1=0
HE-SU_MCS2=0
HE-SU_MCS3=0
HE-SU_MCS4=0
HE-SU_MCS5=0
HE-SU_MCS6=0
HE-SU_MCS7=0
HE-SU_MCS8=0
HE-SU_MCS9=0
HE-SU_MCS10=0
HE-SU_MCS11=0
g_6M_n_6_5M_40M=0
g_9M_40M=0
g_12M_n_13M_40M=0
g_18M_n_19_5M_40M=0
g_24M_n_26M_40M=0
g_36M_n_39M_40M=0
g_48M_n_52M_40M=0
g_54M_n_58_5M_40M=0
n_65M_40M=0
HE-SU_MCS0_40M=0
HE-SU_MCS1_40M=0
HE-SU_MCS2_40M=0
HE-SU_MCS3_40M=0
HE-SU_MCS4_40M=0
HE-SU_MCS5_40M=0
HE-SU_MCS6_40M=0
HE-SU_MCS7_40M=0
HE-SU_MCS8_40M=0
HE-SU_MCS9_40M=0
HE-SU_MCS10_40M=0
HE-SU_MCS11_40M=0

delta_gain1:27 delta_gain2:27 delta_gain3:27 dcxo:50
/tmp #
```



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